

Computational Physics: Materials Science

Exercise 7

The 2026 sodium-cluster challenge

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The goal of this exercise is to generate the Na_{76} cluster with the lowest potential energy under the Morse potential of exercise sheet 3 (the Girifalco–Weizer parameters for sodium). This is a *global*-optimisation problem on the potential-energy surface (PES) of a 76-atom cluster, which has an exponentially large number of local minima. The strategy follows the four parts of the sheet: (a) build a physical initial structure and melt it with molecular dynamics (MD); (b) relax both the cold and the molten structure with several ASE local optimisers and compare them; (c) assess how close the resulting local minimum is to the global one; and (d) use temperature explicitly — simulated annealing and basin hopping — to search for the lowest-lying minimum. The Atomic Simulation Environment (ASE) drives the dynamics and the optimisers. An AI assistant (Claude) was used to develop the ASE Morse calculator, the optimisation and global-search drivers and the plotting; the choice of methods, their interpretation and this report are my own.

Implementation

Potential

The Morse potential models the cohesive bonding of the sodium cluster through a short-range repulsion and an exponentially decaying attraction,

$$V(r) = D \left[e^{-2\alpha(r-r_0)} - 2e^{-\alpha(r-r_0)} \right] \quad (r < r_{\text{cut}}), \quad \mathbf{F}_i = - \sum_{j \neq i} \frac{dV}{dr} \frac{\mathbf{r}_{ij}}{r_{ij}}, \quad (1)$$

with $D = 0.06334 \text{ eV}$, $r_0 = 5.336 \text{ \AA}$, $\alpha = 0.58993 \text{ \AA}^{-1}$ and a hard cutoff at $r_{\text{cut}} = 14.0 \text{ \AA}$ (the pair energy is set to zero beyond r_{cut} ; no shift is applied, matching the literal definition on the sheet). The minimum of a single pair lies at r_0 with depth D , i.e. $V(r_0) = -D$. The dimensionless range parameter is $\rho_0 = \alpha r_0 = 3.148$, a *moderate* range for which the global minima of Morse clusters tend to be compact, polytetrahedral/icosahedral packings rather than open crystalline fragments.

The potential is implemented as a subclass of `ase.calculators.Calculator` that evaluates Eq. (1) for all pairs within the cutoff (a vectorised NumPy kernel, with an optional numba twin for speed; for $N = 76$ the cost is negligible) and writes energy and forces into the `results` dictionary, so the ASE integrators and optimisers use exactly this potential. The analytic forces were verified against a central finite-difference of the energy (maximum deviation $\sim 10^{-10} \text{ eV/}$), and the single-pair check returns $V(r_0) = -D$ exactly. The cluster is placed at the centre of a $60 \text{ \AA}$ cubic box with periodic boundaries switched off (`pbcs=False`); there is therefore no periodic image at all, and the vacuum requirement of the sheet is satisfied trivially (the final cluster spans $\approx 21.6 \text{ \AA}$, leaving a gap of $\approx 19 \text{ \AA}$ on each side, well beyond r_{cut}).

Initial structures and search methods

Three seed geometries are used: a *face-centred-cubic (fcc) sphere* (the 76 atoms closest to the centre of an fcc lattice with nearest-neighbour spacing r_0 , the most compact ordered seed); a *Mackay icosahedron* (the 76 innermost atoms of a four-shell icosahedron); and a *random compact* sphere. The building blocks of the search are:



- **Robust relaxation.** Every quench is a *single continuous* FIRE run to $f_{\max} < 1 \times 10^{-4} \text{ eV/\AA}$. A two-stage chain (or re-relaxing a loosely-converged structure) was found to let momentum carry the cluster over a small ridge into a shallower neighbouring basin, so a continuous run is used so that every minimum compared during the search — and the handed-in structure — is a genuine local minimum.
- **MD heating (a).** A Langevin thermostat ramps the temperature in stages so that the cold seed first relaxes and then melts.
- **Local optimisers (b).** Five ASE optimisers — BFGS, LBFGS, FIRE, MDMin and GoodOldQuasiNewton — are run from both the cold seed and a molten snapshot, recording the energy and the maximum force at every step. Note that the convergence criterion used here is the looser $f_{\max} < 1 \times 10^{-3} \text{ eV/\AA}$, chosen so that step counts are comparable across optimisers; all global-search quenches use the tighter $f_{\max} < 1 \times 10^{-4} \text{ eV/\AA}$ to guarantee genuine local minima.
- **Minima ensemble (c).** Twelve independent relaxations from random starts give a distribution of local-minimum energies.
- **Global search (d).** Simulated annealing (melt then slowly quench, several restarts) and basin hopping. Annealing heats to just above melting ($T_{\text{hi}} \approx 750 \text{ K}$, below the evaporation regime of this weakly-bound cluster) and recollects any atom that drifts off the surface, so the molten cluster cannot boil away. The basin-hopping move set combines a Cartesian “shake” with a *worst-atom relocation* move (the most weakly bound atom is moved to a surface gap), and new minima are accepted by a Metropolis criterion.

Simulation parameters

N	D [eV]	r_0 [Å]	α [Å ⁻¹]	r_{cut} [Å]	$\rho_0 = \alpha r_0$	box [Å]	Δt [fs]
76	0.06334	5.336	0.58993	14.0	3.148	60	5

Table 1: Potential and simulation parameters. The MD uses a Langevin thermostat ($\gamma = 0.01 \text{ fs}^{-1}$); local relaxations converge to a maximum force $f_{\max} < 1 \times 10^{-4} \text{ eV/\AA}$ (a single continuous FIRE run, so every minimum is genuine). The global search uses 12 random-start minimisations, 6 annealing restarts and 250 basin-hopping steps.

Results

(a) Initial structure and MD melting

The cold fcc seed has an (unrelaxed) energy of -30.71 eV (Fig. 1a); its atoms sit on a regular lattice with a high-coordination core and low-coordination corners. Heating it with the Langevin thermostat (Fig. 2) shows two clearly separated stages. At low temperature the potential energy first *drops* to $\approx -36 \text{ eV}$ as the lattice seed relaxes and contracts into a more compact arrangement. As the temperature is then ramped towards $\approx 700 \text{ K}$, the energy rises back above the cold-seed level and fluctuates around $\approx -28 \text{ eV}$: the cluster has melted. The molten snapshot at $T \approx 689 \text{ K}$ (Fig. 1b) is visibly disordered, occupies a larger volume than the seed, and contains both very high- and very low-coordinated atoms — the structural disorder that lets the system escape the ordered seed basin.

Relaxing each seed to a true minimum already separates the candidate motifs: the fcc sphere relaxes to -37.31 eV , the random compact sphere to -37.55 eV , and the Mackay



icosahedron to -37.51 eV. Both the icosahedron and the random quench relax markedly *deeper* than the fcc fragment, so a compact icosahedral/polytetrahedral packing — not the fcc crystallite — is favoured at this range parameter. This is an early hint that the low-energy structures are compact but not crystalline, and that the ordered fcc seed is a comparatively poor starting point.

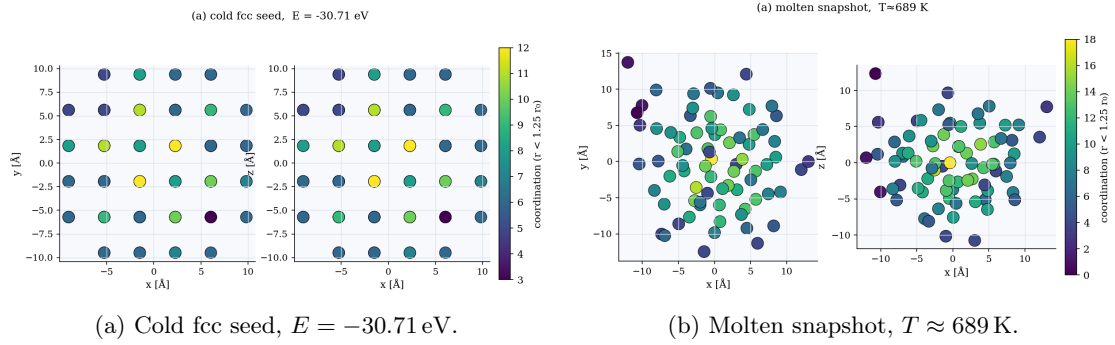


Figure 1: (a) Initial fcc-sphere seed (left) and a snapshot after MD heating (right), both shown in two projections and coloured by coordination number (neighbours within $1.25 r_0$). The ordered seed melts into a disordered, more extended cluster.

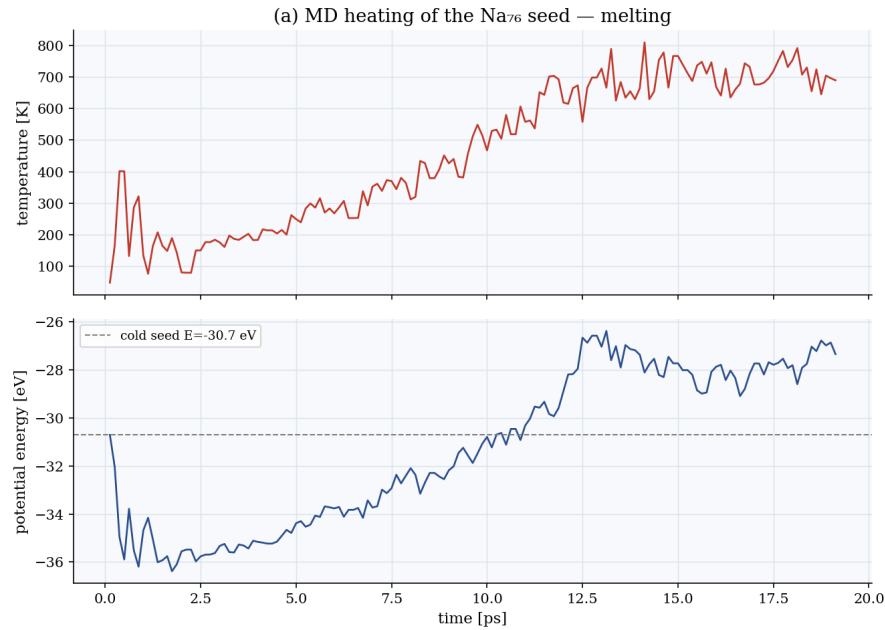


Figure 2: (a) MD heating of the Na_{76} seed. Top: instantaneous temperature ramped from ≈ 50 K to ≈ 700 K. Bottom: potential energy. The energy first drops to ≈ -36 eV as the cold seed relaxes, then rises above the cold-seed reference (dashed, -30.7 eV) as the cluster melts.

(b) Local optimisers and how they differ

Figure 3 compares the five optimisers, started once from the cold seed (left) and once from the molten snapshot (right); the lower panels show the maximum force on a logarithmic axis with the convergence criterion $f_{\text{max}} = 1 \times 10^{-3}$ eV/Å marked.

From the ordered seed. BFGS, FIRE, MDMin and GoodOldQuasiNewton all relax the seed quickly into the same basin at ≈ -37.3 eV, but with markedly different cost: the



damped-MD method FIRE reaches the criterion in the fewest steps, followed by MDMin and BFGS, while GoodOldQuasiNewton converges in energy yet stalls in force and does not reach $f_{\max}$ within the step budget. LBFGS is the outlier: started from the ordered, soft-mode seed it becomes *unstable*, ending at ≈ -29.9 eV with its energy oscillating in a band around -30 eV and its force never falling below $\approx 8 \times 10^{-2}$ eV/Å. This illustrates the central difference between the methods: the quasi-Newton optimisers (BFGS, LBFGS, GoodOldQuasiNewton) build an approximate Hessian from successive forces and take near-optimal steps once the PES is locally quadratic, but a poor initial Hessian estimate can send them off course; the damped-MD methods (FIRE, MDMin) follow Newtonian dynamics with friction and are far more robust on rough or near-degenerate starting points.

From the molten snapshot. Here the outcome is optimiser-dependent. FIRE descends to the deepest basin, -37.58 eV — about 0.27 eV below the best cold-seed relaxation (-37.31 eV) — while BFGS, LBFGS and MDMin all land in a shallower basin near -37.45 eV and GoodOldQuasiNewton is again capped at -37.30 eV. FIRE also converges in the fewest force evaluations; LBFGS, which failed outright on the ordered seed, now converges but is among the slowest. Two physical observations follow: relaxing a *disordered* structure can reach a lower minimum than relaxing the ordered seed (FIRE: -37.58 eV vs. -37.31 eV), and *which* basin is reached depends on the optimiser, because each only finds the basin downhill of its trajectory. Both points motivate searching from many configurations and with temperature, in parts (c) and (d). Table 2 summarises the runs.

Figure 4 shows how the atomic positions themselves evolve during FIRE relaxation from each start. From the cold ordered seed (top row) the atoms barely move: the lattice contracts slightly and corner atoms rearrange, but the overall fcc-like arrangement is preserved throughout, consistent with all quasi-Newton methods landing in the same shallow basin. From the molten disordered snapshot (bottom row) the structural change is dramatic: atoms that begin in random, low-coordination configurations migrate substantially over the first tens of steps and arrive at a compact, high-coordination arrangement that is inaccessible from the ordered seed. This directly illustrates why starting from a melted structure gives access to deeper basins.

optimiser	cold fcc seed ($E_0 = -30.71$ eV)		molten snapshot ($E_0 = -27.34$ eV)	
	E [eV] / steps	converged	E [eV] / steps	converged
BFGS	-37.31 / 271	yes	-37.45 / 875	yes
LBFGS	-29.85 / 1500	no	-37.45 / 1485	yes
FIRE	-37.31 / 138	yes	-37.58 / 500	yes
MDMin	-37.31 / 227	yes	-37.45 / 1396	yes
GoodOldQuasiNewton	-37.28 / 401	capped	-37.30 / 401	capped

Table 2: Local-optimiser comparison (convergence criterion $f_{\max} = 1 \times 10^{-3}$ eV/Å; GoodOldQuasiNewton and the cold-seed LBFGS were stopped at the step cap). FIRE is fastest from both starts and reaches the deepest basin from the molten one. Walltime is dominated by GoodOldQuasiNewton (6–12 s, full-Hessian cost per step) versus FIRE (0.1–0.3 s); LBFGS diverges on the ordered seed.



(b) Local optimizers: convergence from ordered vs disordered start

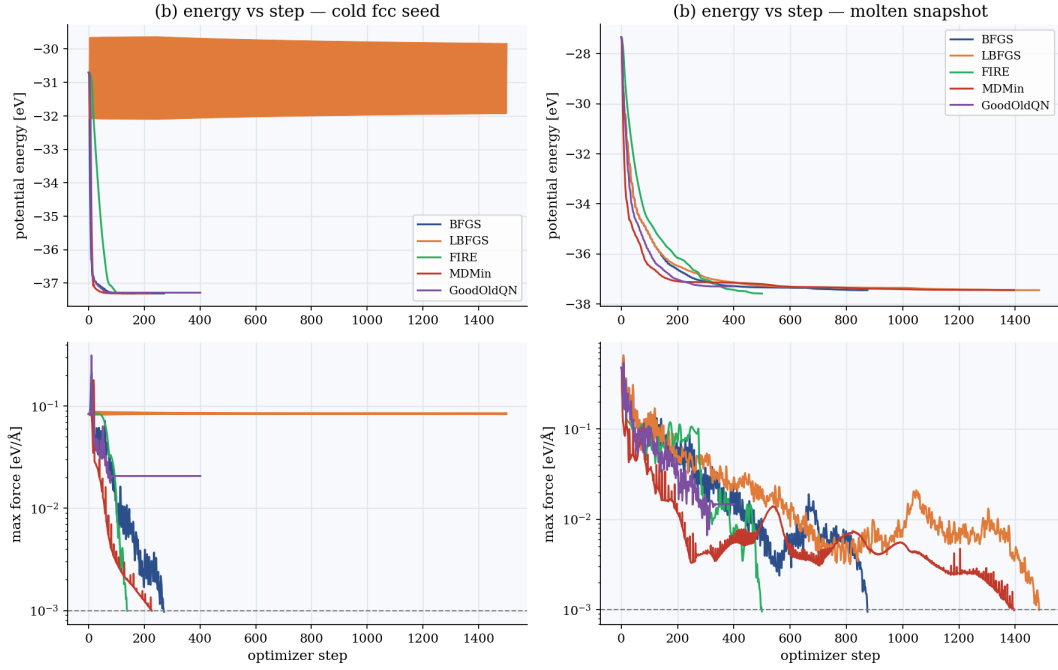
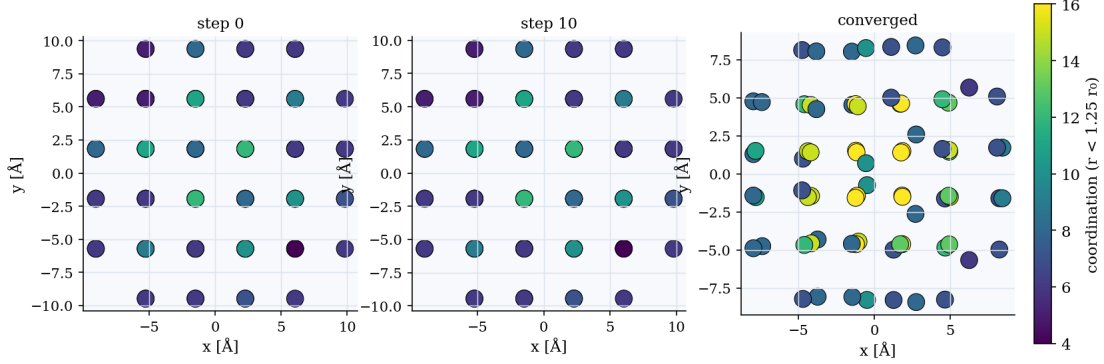


Figure 3: (b) Local optimisers started from the cold fcc seed (left) and from a molten snapshot (right). Top: potential energy vs. optimiser step. Bottom: maximum force (log axis; dashed line = 1×10^{-3} eV/Å). From the ordered seed LBFGS (orange) fails to converge while the others reach ≈ -37.3 eV; from the molten start FIRE reaches the deepest basin (-37.58 eV) and the others land near -37.45 eV (GoodOldQuasiNewton capped at -37.30 eV).

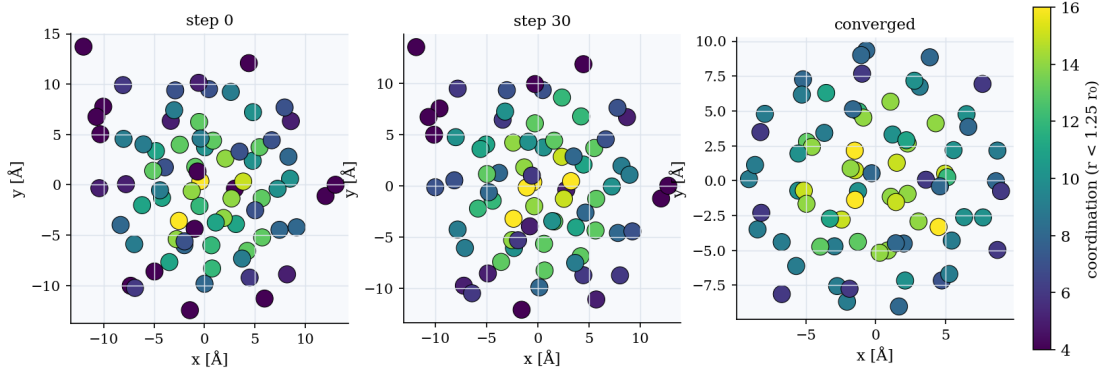


(b) FIRE trajectory snapshots — cold fcc seed



(a) FIRE from the cold fcc seed: atoms remain near their lattice positions throughout.

(b) FIRE trajectory snapshots — molten snapshot



(b) FIRE from the molten snapshot: large-scale atomic rearrangement produces a compact, high-coordination structure inaccessible from the ordered seed.

Figure 4: (b) FIRE trajectory snapshots (xy-projection, coloured by coordination number): atomic positions at the start, an intermediate step, and the converged minimum. Each panel illustrates how much structural change the optimiser drives during relaxation.

(c) Is the local minimum close to the global minimum?

A single relaxation cannot by itself certify a global minimum. To gauge how close we are, twelve independent relaxations were run from random starting structures and their final energies histogrammed (Fig. 5). They spread over a band of competing minima from ≈ -37.38 eV to -37.627 eV; the lowest random quench, -37.627 eV, still lies about 52 meV *above* the global minimum located in part (d) (-37.6794 eV). This is the key message: for a 76-atom cluster the potential-energy surface has so many basins that plain random quenching does *not* reach the global minimum — it samples the dense cloud of shallow minima and essentially never lands in the deepest basin. A dedicated global search is therefore necessary, not optional.

Confidence that the -37.6794 eV structure is at (or very near) the global minimum then rests on (i) multiple independent basin-hopping chains converging to the *same* energy and structure; (ii) that energy lying well below the entire random-quench cloud (Fig. 5); and (iii) its consistency with the compact, high-coordination motif expected for a moderate-



range Morse cluster. Increasing confidence further would mean more independent chains, finer search moves, and comparison against tabulated Morse-cluster global minima at this range parameter. A rigorous proof of globality is not possible by search alone.

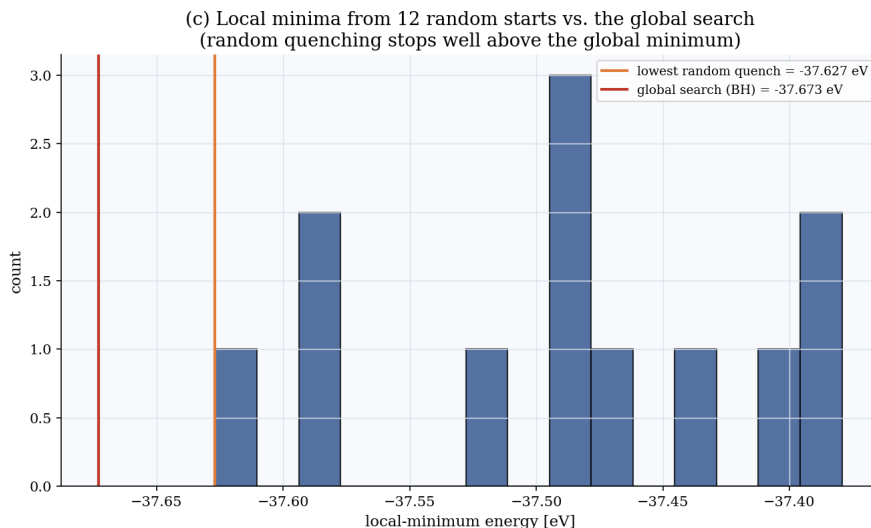


Figure 5: (c) Local-minimum energies from twelve independent random-start relaxations (each a continuous tight relaxation to a true minimum). The lowest random quench (orange, -37.627 eV) lies ≈ 52 meV above the global-search result (red, -37.6794 eV): random quenching stops well above the global minimum.

(d) Using temperature: annealing and basin hopping

The strategy that uses temperature explicitly is *simulated annealing*: heat the cluster until it is liquid so it can rearrange over barriers, then cool it slowly so it settles into a low-energy basin. Figure 6a shows the potential-energy traces of six melt-quench restarts. Each is first relaxed to remove bad contacts, then heated to ≈ 750 K where the energy rises to ≈ -27 eV (the molten plateau), and finally cooled in stages back down to ≈ -37 eV. Relaxing the cooled structures gives compact minima reaching as low as -37.64 eV, already deeper than the random-quench ensemble of part (c). Because the cluster is only weakly bound, the melt temperature must be kept below the evaporation regime and stray atoms recollected, otherwise atoms boil off and the anneal fails; with that control the six restarts cool reproducibly into low-energy basins. A single anneal still does not reliably hit the *global* minimum, which is why several restarts and a basin-hopping stage are used. **In a real experiment this is literally thermal annealing** — a sample is heated to let its atoms diffuse and reorganise, then cooled slowly so it relaxes towards its ground-state structure instead of freezing into a defective one.

Basin hopping refines the search by repeatedly perturbing the current structure (a Cartesian shake or a worst-atom relocation), relaxing to the nearest true minimum, and accepting or rejecting by a Metropolis rule. Figure 6b shows a single chain started from the shallow fcc minimum (-37.31 eV): the best energy so far falls in distinct steps to -37.673 eV within 250 hops — already within a few meV of the global minimum. Running several such chains (an extended basin-hopping search) locates the deepest structure, -37.6794 eV (dashed line). The relocation move (moving the most weakly bound atom to a surface gap) produces the largest single improvements, because it lets the search

cross between structural motifs that a small Cartesian shake cannot reach.

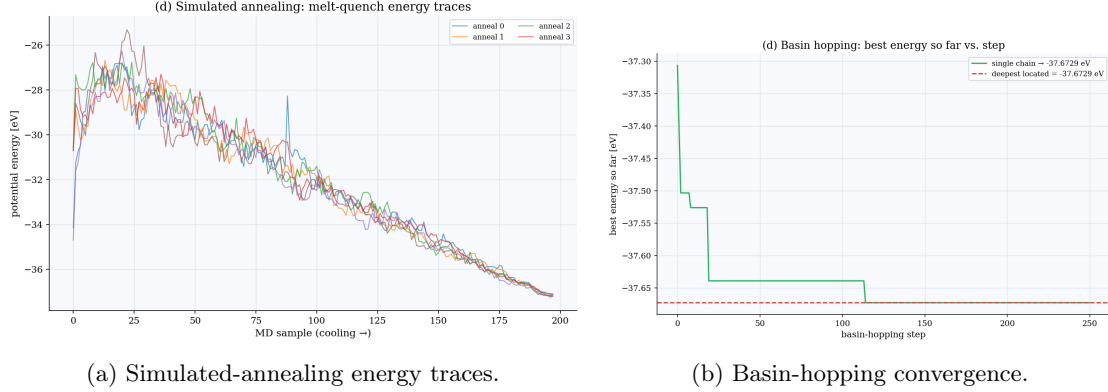


Figure 6: (d) Temperature-driven global search. (a) Six melt-quench restarts heat to a molten plateau (≈ -27 eV) and cool reproducibly to ≈ -37 eV. (b) A single basin-hopping chain from the fcc minimum falls step-wise to -37.673 eV; the deepest structure located by extended basin hopping is -37.6794 eV (dashed).

The winning cluster

The lowest configuration located by the search is shown in Fig. 7. Every relaxation in the search is a single continuous FIRE run to a true minimum (no optimiser restart, which would let a loosely-converged structure slip into a shallower neighbouring basin), so the handed-in structure is a genuine minimum at $f_{\max} < 1 \times 10^{-4}$ eV/Å:

$$E = -37.6794 \text{ eV} = -0.4958 \text{ eV/atom} = -7.83 D, \quad (2)$$

i.e. a cohesive energy of 7.83 pair-well depths per atom. The cluster is compact and roughly spherical, with a high-coordination core (up to ≈ 16 neighbours within $1.25 r_0$) and a low-coordination surface (down to ≈ 6), as expected for a moderate-range Morse cluster. It is saved as `Na76_cluster_StefanusMichael.traj`.

WINNER $E = -37.6729 \text{ eV}$ (-0.4957 eV/atom)

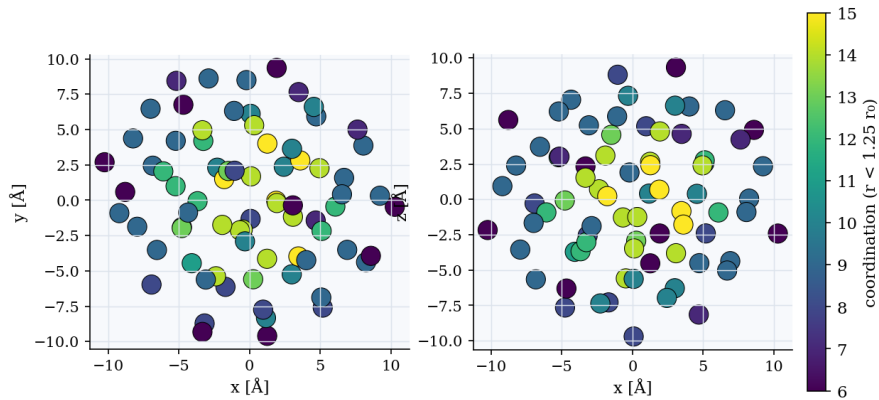


Figure 7: Lowest-energy Na_{76} cluster, $E = -37.6794 \text{ eV}$ (-0.4958 eV/atom), shown in two projections and coloured by coordination number. The dense, high-coordination core is surrounded by a lower-coordination surface.

Figure 8 compares the lowest cluster from three different methods. Local optimisation of the fcc seed retains visible lattice ordering and stops at -37.31 eV ; simulated annealing



and basin hopping both produce compact, disordered structures, reaching -37.64 eV and -37.68 eV respectively. The progression — ordered/shallow $\rightarrow$ disordered/deeper — is the visual summary of the whole search.

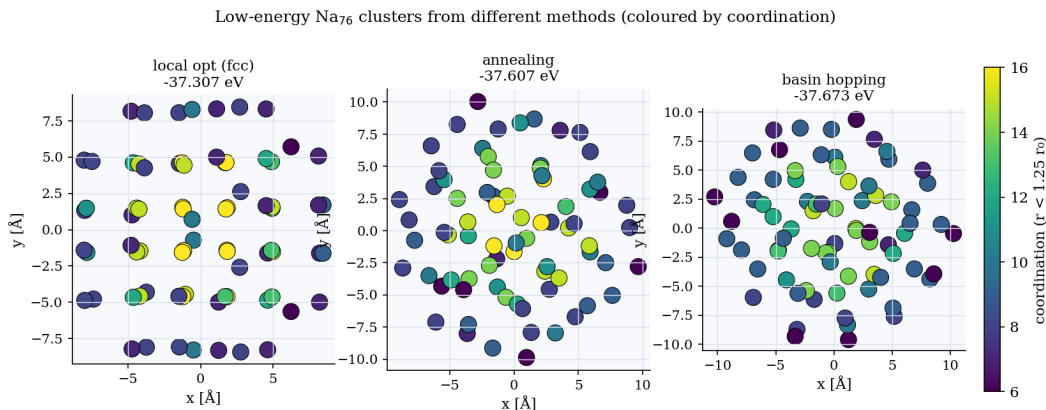


Figure 8: Low-energy Na_{76} clusters from different methods (one projection each, coloured by coordination): local optimisation of the fcc seed, the best simulated-annealing minimum, and the basin-hopping winner. The ordered fcc relaxation is the shallowest; the disordered annealed and basin-hopped structures are deeper.

Key takeaways

- **Locating local minima.** Two families were used. *Quasi-Newton* methods (BFGS, LBFGS, GoodOldQuasiNewton) approximate the Hessian from successive forces and converge fast near a quadratic minimum, but are sensitive to the starting point (LBFGS failed on the ordered seed). *Damped-MD* methods (FIRE, MDMin) integrate Newton’s equations with friction; they are slower in the quadratic region but much more robust on disordered starts — FIRE was the most reliable and fastest overall here.
- **Searching for the global minimum.** A single relaxation only finds the nearest basin. The harder global problem is tackled by sampling many starts (c) and by hopping between basins with temperature (d): simulated annealing (melt then quench slowly) and basin hopping with a relocation move. Confidence that the result is global comes from independent methods agreeing on the same energy and structure, not from a single run.
- **Applications of geometry optimisation.** Beyond this challenge, relaxing a structure yields equilibrium geometries and bond lengths, cohesive and defect-formation energies, and — via the Hessian at the minimum — vibrational frequencies and elastic/thermodynamic properties; transition-state searches between minima give reaction barriers. The optimised structures are also the physical starting points for further MD or property calculations.

Summary

- **Result.** The lowest Na_{76} Morse cluster found has $E = -37.6794$ eV (-0.4958 eV/atom, $-7.83 D$), a compact roughly spherical structure with a high-coordination core.
- **Potential.** A custom truncated Morse ASE calculator ($\rho_0 = 3.148$), verified against finite-difference forces and $V(r_0) = -D$; the cluster sits in vacuum (`pbc=False`,



$\approx 19 \text{ \AA}$ gap $> r_{\text{cut}}$).

- **Seeds and melting (a).** Relaxed to true minima, the icosahedron (-37.51 eV) and a random quench (-37.55 eV) both go *deeper* than the fcc seed (-37.31 eV), so compact icosahedral/polytetrahedral packing is favoured over the fcc fragment. The fcc seed (-30.71 eV unrelaxed) melts above $\approx 700 \text{ K}$ into a disordered cluster.
- **Optimisers (b).** From the cold seed all methods reach $\approx -37.3 \text{ eV}$ except LBFGS, which diverges; from the molten start the outcome is optimiser-dependent — FIRE reaches the deepest basin (-37.58 eV) while BFGS/LBFGS/MDMin land near -37.45 eV and GoodOldQuasiNewton at -37.30 eV . FIRE is fastest and most robust throughout.
- **Global search (c,d).** Twelve random quenches stop between -37.38 eV and -37.627 eV , none reaching the global minimum — random quenching alone is insufficient. Simulated annealing (best -37.64 eV) and basin hopping (with a worst-atom relocation move) locate the deepest structure, -37.6794 eV . Annealing is the simulation analogue of thermal annealing in a real experiment.

